

Theatre Management System

1. PROBLEM STATEMENT

Traditional box-office and movie ticketing setups often face inefficiencies due to manual record-keeping, seat tracking errors, and slow billing processes. Without a centralized system, movie theatre administrators and ticket vendors struggle with real-time seat availability tracking, preventing double-bookings, and quickly generating accurate billing information for patrons.

There is a critical need for a streamlined, command-line management system that simplifies movie browsing, seat selection, real-time status tracking, and automated bill generation to reduce operational overhead at small theater ticket counters.

2. SCOPE OF THE PROJECT

The current scope of the **Theatre Management System** encompasses a lightweight, terminal-based Python application designed to manage key cinema operational workflows:

- Movie Catalog Management:** Displaying and retrieving available movie titles dynamically.
- Seat Reservation:** Displaying real-time availability for a fixed set of seat inventory (10 seats) and preventing duplicate bookings.
- Billing & Receipt Generation:** Automatically calculating total pricing based on selected tickets and generating detailed billing summaries.

3. TARGET USERS

Front-Desk Staff / Operators	Independent Theatre Owners	Students & Developers
Terminal operators who interact with customers in person to check showtimes, select seats, and process ticket orders quickly.	Operators looking for a simple, minimal-overhead software solution to manage small-scale movie screening inventory.	Individuals learning procedural/modular Python design and basic CLI state management principles.

4. HIGH-LEVEL FEATURES SUMMARY

Module	Primary Component	Key Functionality
Movie Catalog	<code>movies.py</code>	Displays structured list of movies; handles dynamic title selection.
Seat Management	<code>booking.py</code>	Tracks real-time status (Available vs. Booked); validates selections.
Billing Engine	<code>billing.py</code>	Calculates totals (\$200/ticket); outputs complete itemized receipts.
CLI Navigation	<code>main.py</code>	Menu-driven system orchestrating user interaction and workflow execution.

